

CrickIQ: An Explainable AI-Based Cricket Analytics and Decision Support System for T20 International Cricket

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Abstract—A considerable amount of data is generated by cricket. Each ball bowled is added to valuable records of runs scored, wickets taken, venues used, behavior of the pitch, and situation in the match. Those who are tasked with collecting data, especially coaches and analysts who are responsible for determining the lineup and strategy, do so from distinct sources, and sometimes that can complicate the decision-making. This paper presents CrickIQ, a decision-support web-based platform for cricket matches. CrickIQ integrates data about players' and teams' performance, venue characteristics, pitch conditions, opponent statistics, forming the ideal squad, forecasting matches, match management, tournament management, player training, and a chatbot which can answer questions in plain language. The lineup module ranks players based on their batting, balling, field contribution, role in the game, fit for conditions, ability to play at the venue, and efficiency against the opposition. The analysis of the pitch includes its composition, wetness, hardness, cracks, bounce, weather conditions, etc. The module providing details regarding competitors enables the user to identify the key players of the opposing team and concentrate on them during preparation. The app was developed using React, TypeScript, TanStack Start, Tailwind CSS, Recharts, and Supabase, and comprises functions for authentication and access permissions based on user roles in the system. There is a single design principle followed by the software, i.e., the system supplies information regarding the actions and recommendations; however, the decision-making always rests with the person.

Index Terms—Cricket Analytics, Decision Support, Player Evaluation, Playing XI Recommendation, Rule-Based Intelligence, Explainable Decision Support

I. INTRODUCTION

When you ask a cricket coach about the experience on the eve of a cricket match, you might not get a calm answer. There are teams to analyze, conditions to assess and rivals to study

all within a very short amount of time. Cricket is a sport which has a lot of data. In fact, just one game produces data about all the bowlers and batsmen in it, as well as information about the venue, the pitch and the details on the incident[1].

The emergence of T20 cricket has made this data even more precious. Games are short, differences between teams are minimal and a single great over or a dropped catch can determine the winner. When it comes to such quick formats, preparation makes a bigger difference than in longer formats.

Traditionally, selection and strategic planning were based solely on the experience and feeling of coaches and specialists. While that experience remains extremely useful today, nothing in this article argues against this statement. However, today's professional teams have plenty of data to work with, which readily complicates the situation. The problem is that there is enough data. The current problem lies in the fact that the data is divided into different sources. The numbers documenting a player can be contained in one database, the stats concerning the stadium in the second one, the results of the match in the third one, and the analysis of the opponents in the fourth one. As a result, it is necessary to get access to all databases, process them, compare them, and summarize all data for making a decision.

Think about what questions a coach has to answer before one match: Are the players performing well at this time?, Is any of the players playing well under the conditions at this ground?, Which players would pose difficulty for his cricket team?, Would the wickets favor spinners or fast bowlers?, Is the provided XI able to perform actions equal to the match strategy, namely, to be properly balanced taking into account many aspects in play?, Is the opponent side strong enough,

and who would hurt us more in this situation?

All these questions are, of course, relevant. However, answering them manually under time pressure is hard, moreover, it causes inconsistencies because two people evaluating the data from the same scatter information can get totally different answers only because of different choice of pieces of information. Artificial intelligence and data analysis are created for the coach's benefit. Hence, coaches do not need to spend too much time thinking about collecting information and can concentrate on decision making instead. The decision support system should be designed correctly so that it gathers all relevant data and organizes it properly together with providing analytical recommendations that can be followed, altered, or ignored.

This research paper reviews a platform called CrickIQ. This platform is T20 cricket-oriented and makes it possible to analyze many features of the game such as player and team analysis, pitch and venue intelligence, opponent analysis, match management, live scoring during the plays, and tournaments managing. Moreover, there is an AI assistant that gives advice in a simple manner.

II. LITERATURE SURVEY

The concept of data utilization in decision making is well known. In their work, Davenport and Harris [2] argued that businesses that systematically use analytics rather than relying on it sporadically are able to build a competitive advantage around the decision-making processes they have. The focus of the authors lies in the fact that the analytics is really important when incorporated into the decision-making process rather than reserved for the specific occasions. The issue is crucial in cricket coaching. A coach should not be provided with the analytical reports on a seasonal basis but should get relevant information regularly.

Data mining and statistical procedures. The essence of transforming data into forecasting is provided in numerous sources on data mining and statistical learning [3]. Their authors describe how to prepare the data, select the model, evaluate its efficiency, and avoid common mistakes, such as overfitting. Although the methods are not related to sports, they can be easily applied for sports data processing. There is an important point to remember: a model's effectiveness depends on its input data. This is important for CrickIQ because the quality of yields is determined by the comprehensiveness of its cricket data. Use of ensemble and boosting methods—many algorithms have been proposed for tables like the one derived from cricket. Breiman's random forests [4] construct multiple decision trees to provide stable predictions, thus preventing overfitting and revealing the most important input values. XGBoost [5] invented by Tianqi Chen and Carlos Guestrin is a scalable version of the gradient boosting trees algorithm and has become very popular in the area of structured data analysis. So the two approaches can find application in cricket in predicting match results and in evaluating players' performances. Note that for now CrickIQ is using a simpler transparent scoring technique; nonetheless

these approaches may be employed in future. Cricket data and rules. The International Cricket Council publishes its rules for international T20 games and has its statistical rankings of players and teams [6]. These sources delineate the type of format as well as provide information that serves as the base figures to the cricket analysis tool being created. They provide the information on what has happened in this sport and what the rule is but not provide the coach with directions on what they should do in this situation. Filling in this gap is what CrickIQ is trying to achieve. Web portals and tools that are used for development. In terms of the technical aspect, today's web frameworks and services mean it is possible for a little team to create the whole analytical application. The documentation that can be found in React [7] and Supabase [8] explains the component-based interface model that CrickIQ is based on and managed services in Supabase. It is important to make the right choice here because if the analytical tool is slow or difficult to use, it will not be used regardless of its reliability and effectiveness.

The literature combined provides the following: justification for analytics-based decision-making, a number of tools ranging from statistical techniques to machine learning approaches, as well as reliable reference points in cricket data and rules. However, it does not provide a simple practical solution that can implement these ideas into day-to-day/contextual activities that need to be undertaken by T20 cricket coaches, taking into account pitch conditions, location history and competing side problems.

III. EXISTING SYSTEM AND PROBLEM IDENTIFICATION

Currently, there are basically three categories of cricket evaluation that can be defined. They include manual evaluation utilizing scorecards, computing devices as well as anecdotal information; cumulative presentation of previous data from publicly available sources or commercial tools, and evaluation applications which apply either automatic learning or conventional statistics. Each method has advantages and disadvantages, but together they demonstrate necessity to create better options for director and coaching choice. Four issues can be named right now. First, selection assessment and evaluation parameters are applied in different ways which makes it impossible to understand why one player is picked instead of another. Second, situational aspects are rarely taken into consideration during selection process, as they are employed afterwards intentionally. Thirdly, one can hardly find a solution to the problem with improper application of the majority of positional players calculations while creating a team. Last but not least, useful functions are spread on different platforms which requires constant movement of information among devices. Thus, a clear necessity in such a complex platform which collects players evaluation, situational evaluation, selection, management and reporting appears. In addition, such a platform needs to provide explanations of its recommendations; it must be possible to connect each piece of advice to the underlying statistics, principles, and conditions. Only at this

point can coaches and analysts have faith in such an output while making the final decisions themselves.

IV. PROPOSED METHODOLOGY

CrickIQ employs a comprehensive workflow process. Its data goes through a data-gathering stage where it assesses the players, teams, venues, matches, ball-by-ball events, phase-based statistics information. This information is gathered from Supabase or local CSV files depending on whether the database is available. The information is then processed, converted to numerical formats, and then used to create metrics before delivering it to the analytical engine. The engine consists of a set of modules, which use mathematical equations, heuristics, and rules making it possible to trace the outputs back to the various inputs from which they generate, making it a valuable decision-making tool.

A. Data Collection

The prototype we are currently using is now equipped with local datasets that include data about teams and players, stadiums, matches, ball-by-ball records, phase statistics as mentioned in Table I. These datasets will perform the function of demonstrating how the system works, however, it needs to be emphasized that the datasets are a small portion of the total cricket records. The system does not connect to a verified live cricket data API, meaning that even though a Supabase database is available, there is no guarantee of receiving constantly up to date data from an external source. The dataset might be complemented by a variety of open datasets in the future, like the Asia Cup 2025 T20 dataset among others.

B. Preprocessing and Derived Statistics

The fields are taken out and converted into number formats, and problems related to improper data entries are corrected before the process of analysis begins. Then, the next step is calculation of derived statistics based on primary statistics: the strike rate is computed using the formula runs scored divided by balls faced multiplied by 100, the economy rate is calculated by number of runs given divided by number of overs bowled, while the bowling average is obtained by calculating the runs given divided by the number of wickets fallen.

C. Player Impact Score

The player-impact mechanism combines batting, bowling and fielding contributions into one composite score. Conceptually, for player i ,

$$I_i = w_b B_i + w_w W_i + w_f F_i \quad (1)$$

where B_i , W_i and F_i are the batting, bowling and fielding contributions and w_b , w_w and w_f are fixed weights defined in the application. The weights are set by design and are not learned from data.

TABLE I
DATASETS USED IN CRICKIQ

Dataset	Contents	Records
Players	Role, styles, career statistics	69
Teams	Squads, ratings	8
Venues	Pitch type, scoring trends	12
Matches	Teams, venue, toss, result	6
Ball-by-ball	Delivery-level events	10
Phase statistics	Powerplay, middle, death	101

D. Pitch and Condition Analysis

Pitch analysis involves explicit characteristics like the amount of grass, moisture, hardness, cracks, and bounce which are to be used as input for tactical calculations. For example, if there is more grass cover and moisture, seam bowling can be effective, hard surface and good bounce will favour pace and stroke play, and a dry worn-out surface will favour spin. The rules are visible and open to inspection so coaches can understand the reasons making tactical recommendations.

E. Playing XI Selection

The proposal for the Playing XI system refers to a smart method constrained by requirements. The players are categorized based on their performance, and the program continuously selects the best player available at each point in time who satisfies the required conditions, namely, the minimum number of batsmen required, wicket-keeper, pacers and spinners, until 11 players are ready.

$$p^* = \arg \max I_p, p \text{ in } A \quad (2)$$

Where A is the collection of leftover players for whom their part is still required. This method is simple and easy to describe but it does not ensure an optimal team in terms of outcome.

F. Opponent and Development Analysis

Opponent analysis modifies player assessment for the opposing side and points out matchups. Player-development analysis compares phase-wise statistics with predefined cut-offs for the purpose of identifying strengths and weaknesses in certain game phases.

G. Conversational Assistance

A local aide uses keywords, patterns and intent identification in order to provide answers to questions concerning players, teams and pitches. To use AI assistance allows getting generative responses. This component is autonomous; it doesn't use the analytical engine to calculate scores or recommendations

V. SYSTEM DESIGN AND IMPLEMENTATION

A. Architecture Overview

CrickIQ makes use of a four-layer architecture that has a presentation layer, application layer, analytical layer, and the persistence layer.

The presentation layer is implemented using React along with TypeScript and is utilized for the user interface. The styling and UI components are implemented by way of the Tailwind CSS and Radix UI. The data is visualized using Recharts.

The second layer of the four-layer architecture is the application layer which has been implemented using TanStack Router for routing as well as TanStack Query for data fetching and caching process.

The analytical layer consists of modules regarding player impact scoring, pitch analysis, Playing XI selection, opponent analysis as well as match recommendations.

The persistence layer has the responsibility of using Supabase for storing and managing of data related to the application.

The computing functions take place in the client side as well as in the memory which allows the application to gain analytical results quickly without any cost of extra server requests.

B. Database Design

The database is designed using Supabase with relational constraints, foreign keys, and check constraints. The database consists of several groups of tables responsible for user management, cricket data, and match operations.

1) *Users and User Access:* The profiles and user roles tables store user information and associate users with one of four application roles: manager, coach, analyst, or player.

2) *Cricket Objects:* The teams, players, player stats, and player phase stats tables store information about teams, players, career statistics, and phase-wise performance.

The venues table contains venue-specific information, including average first-innings scores and the relative influence of pace and spin bowling conditions.

3) *Match Control:* The tournaments, matches, match squads, innings, and balls tables manage tournament and match information. These tables store fixtures, selected squads, innings information, and ball-by-ball events.

Row-level security policies are implemented to ensure that users can access only the information permitted by their assigned roles.

C. Player Impact Scoring

The Player Impact Scoring module evaluates player performance using batting, bowling, and fielding statistics. The module combines relevant performance indicators to produce an overall player impact value.

The scoring mechanism can be applied across T20 match format and provides a common analytical measure for comparing players. The resulting impact score is used by other analytical modules, particularly the Playing XI selection and match recommendation components.

D. Match Suggestion

The Match Suggestion feature utilizes available information on players, teams, venues, and matches to come up with

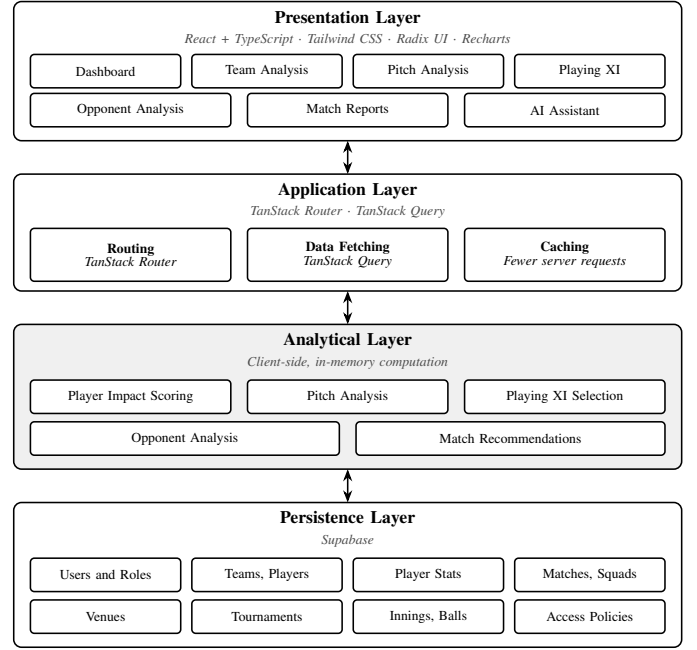


Fig. 1. Four-layer architecture of the proposed CrickIQ platform.

analytical recommendations. The feature also makes use of the past performances and the relevant conditions of the match to come up with.

Calculated metrics can be shown on the interface of the application to help coaches and analysts assess possible strategies for their matches.

E. Pitch Analysis

The Pitch Analysis feature determines weather and pitch conditions in order to evaluate their monitor performance in the field. The analysis takes into consideration variables of the playing surface to figure out whether the conditions are more favorable for pace or spin bowling.

The end result of the pitch analysis is used by other parts of the system, including Playing XI selection and tactical recommendation modules.

F. Playing XI Selection

The Playing XI module uses a constrained greedy selection algorithm to construct a balanced Playing XI. The algorithm fills the team according to predefined player roles and match requirements.

The selection process considers the following roles:

- Two opening batters.
- An anchor at the third position.
- A power hitter at the fourth position.
- A specialist selected according to team requirements.
- A finisher based on death-over performance.
- A wicketkeeper.
- An all-rounder selected according to pitch conditions.
- Specialist bowlers.

The selection criteria require at least one wicketkeeper, a minimum of five bowling options, and at least four specialist batters. The selection of player types is adjusted according to the effectiveness of the pitch for pace and spin.

The greedy approach provides a fast and understandable method for constructing the Playing XI. However, the current approach does not guarantee that the selected team is globally optimal in terms of overall performance.

G. Opponent and Development Analysis

The Opponent Analysis component reviews the statistics of batting and bowling of competing players along with their past records and performance history. The Player Development component grades the performance of players under various circumstances linked to types of bowling and stages of play. It makes it possible for the system to detect the weaknesses and areas that need improvement.

The system detects a player who has the average of fewer than 22 or the strike rate lower than 115 against left-handed medium pace bowlers. All such insights can then be used to development recommendations that will be more effective.

H. Operational Modules and AI Assistant

The system contains several operational modules, including a live scoring console, pitch simulation studio, opponent comparison interface, and report generation module.

The scoring console records individual balls and calculates important match statistics such as current run rate, required run rate, and partnership runs.

The report generation module produces PDF match sheets and team reports using the jsPDF library and generates spreadsheet-based data using SheetJS.

The AI Assistant provides a natural-language interface for querying the analytical engine. It can respond to questions related to death-over bowling options, pitch conditions, squad balance, and player matchups.

The assistant uses intent recognition to identify the type of query and generates responses using values calculated by the analytical engine.

I. Technology Stack

The major technologies used in the implementation of CrickIQ are:

- **Frontend:** React and TypeScript.
- **Styling:** Tailwind CSS.
- **UI Components:** Radix UI.
- **Data Visualization:** Recharts.
- **Routing:** TanStack Router.
- **Data Management:** TanStack Query.
- **Backend and Database:** Supabase.
- **Report Generation:** jsPDF.
- **Spreadsheet Generation:** SheetJS.

VI. RESULTS AND DISCUSSIONS

The CrickIQ platform that has been used is uniquely assembled by combining various functions for the purpose of centralized cricket analysis and decisions making process. The components of data ingestion and preprocessing provide a structured means of bringing in cricket information into the CrickIQ platform. When processing is completed, the important cricket information regarding players, teams, venues, matches and phases is analyzable under the context of role, format, venue, opponents and playing phase.

The function player analytics changes bare performance indicators into a unique understandable impact mark. Batting, bowling and fielding contributions are summed together based on certain weights allowing to judge all players uniformly and explanations for any indexes to be found out.

The pitch analysis module transforms the condition of the surface into strategies. The aspects such as the moisture level in grass, hardness and cracks, bounce, dew and others are regulated by the rules in order to decide if spin or pace is favored and help create a bowling plan. The Playing XI module is a more advanced one as it ranks players according to their influence and takes into account the pitch type as well as the rival team.

The opponent analysis process looks at batting handedness, bowling hand and the performance record between the two players. The player development analysis compares performance in phases including powerplay, middle overs and death overs to overall performance to identify weaknesses and strengths which can be tapped into when training the player. The suggestion for any match integrates team rating and squad strength, advantage of playing at home and contribution of pitch and gives a number to help prepare for the match.

The platform includes not only analytical tools but also management system which allows to manage cricket. The match scoring is suitable both for ball-by-ball and scoreboard and other functions to manage tournament, matches, and teams. With the help of reporting module, coaches and analysts get reports. The team balance are the perfect support for the decision-making process during the squad formation as they allow to assess players' value and find weaknesses in the bowling attack or fielding positions. Now, conversational assistant provides an additional level of interaction. Instead of opening different pages searching for the required information, the user can ask the AI to answer any question that arises. However, the assistant is separated from the analytical engine as it is not able to perform any calculations regarding the impact score, pitch, and selection.

A. Functional Validation

The performance of each component was established based on the task it was created to fulfill. In particular, it needed to be demonstrated how the component performed its job and how it was integrated into the system as a whole. For example, the impact score had to be accurately calculated based on statistics inputted into that component, the Playing XI selector had to follow its constraints, while the pitch component had to

TABLE II
FUNCTIONAL VALIDATION OF THE IMPLEMENTED PLATFORM

Module	Validation Focus
Authentication	Login and role-based access
Data loading	Database, CSV and fallback sources
Player analytics	Impact score calculation
Team analysis	Squad composition and balance
Playing XI	Role-constrained greedy selection
Pitch analysis	Rule-based tactical output
Opponent analysis	Matchup and threat identification
Match suggestion	Logistic probability output
Player development	Phase-wise strengths and weaknesses
Matches and scoring	Match records and live scoring events
Tournaments	Fixtures and standings
Reports	PDF and spreadsheet generation
AI assistant	Local responses

provide the pitch condition recommendations. Authentication, data loading, scoring, tournaments, and reporting components were tested to see whether they performed their tasks.

It is also essential to point out what this validation proves and what it does not. In other words, it proves the functionality of the component without statistics being present. It proves that the components fulfill their task but does not prove how accurate the performance suggestions are regarding matches that the system has never come across before. This kind of accuracy testing needs much broader data and the test itself can be carried out in future studies.

B. Discussion

The main advantage of CrickIQ is that everything in regards to analysis and management takes place on one site. There is no more need for a coach to switch back and forth between statistics sites, spreadsheets and score calculation tools to put the same data. Player performance, pitch condition, matchups with the opposition team and squad balance are analysed in conjunction, with the results generated in one application. The next benefit is transparency. Each result stems from some rule or formula. The coach is able to trace the origin of each advice coming from the system. If a certain player is named as a recommendation, it means that the coach can see its explanation in terms of his batting, bowling or fielding scores or some match-up benefit. This is especially true for coaching, because the chances are that coaches are more likely to listen to some advice they understand rather than to some score which appears from nowhere. The use of the same design presents certain compromises. The deterministic model is unable to learn through data, and therefore the model performance depend solely on how well the algorithms and rules model real cricket phenomena. If there is a small discrepancy in the coefficients, this error will be carried through the model. A machine learning model could capture non-linear relationships in the data that are ignored by the deterministic model, for example, the relationship between form, conditions, and matchups. Figure 2 shows the heatmap

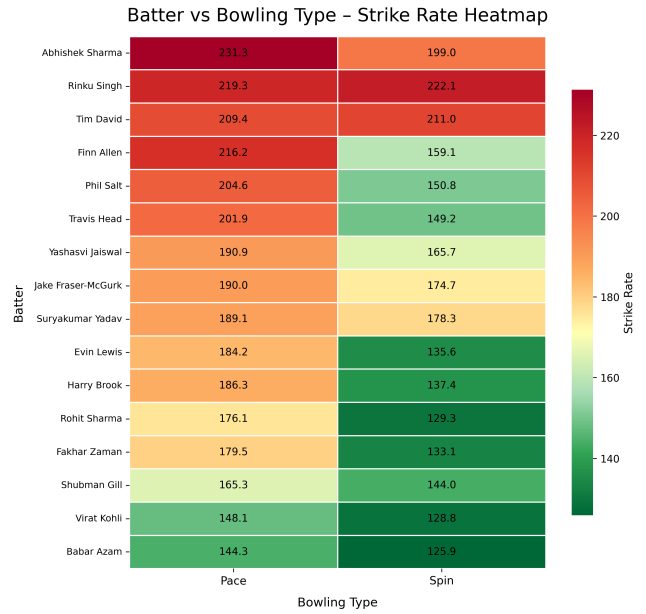


Fig. 2. Batter performance against pace and spin bowling based on strike rate.

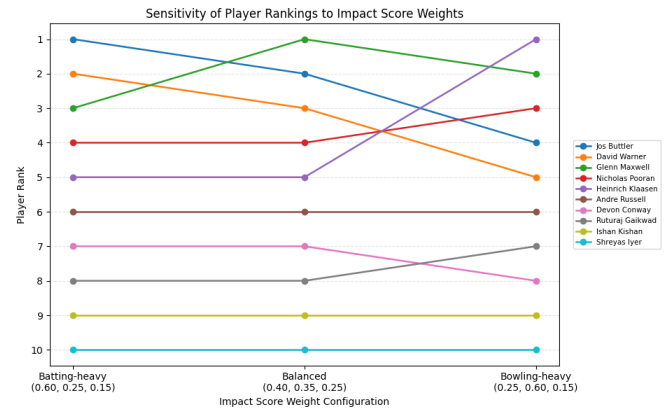


Fig. 3. Sensitivity of player rankings to different Player Impact Score weight configurations.

comparing batter performance against spin and pace bowling based on the strike rate. Figure 3 shows the analysis chnages on how when batters, ballers, fielding infolunces each players ranking.Batting heavy, Balanced and Balling heavy are compared here. However, machine learning comes with its own challenges of explanation, use of huge amounts of data, and careful validation. CrickIQ prioritizes interpretable solutions. A direct comparison of the two systems is a defined research agenda.

C. Limitations

CrickIQ has various constraints at present. It has no trained machine-learning models in place and has never been tested on any previously unobserved matches. Thus the insights generated by the system can only be taken as assistance in decision-making, since they lack verification.

The size of the datasets is relatively small when compared to the total available cricket data, and there is no valid live data feed. Therefore, the results only cover the data loaded into the system and it has to fall back upon formulas and other predetermined mechanisms in certain cases.

The coefficients and weights of impact have been set manually, without any analytical verification of results achieved historically. The algorithm that gives the solution does not guarantee the optimal solution, and therefore, the role of AI is rather limited.

The result in every situation depends on the quality of the input data. And hence CrickIQ may be regarded merely as a prototype, but not anything more serious.

VII. CONCLUSION

CrickIQ is the name given to a cricket decision-making and analytics platform based on the web. This product provides all necessary services for T20 cricket analysis, such as player analysis, team analysis, Playing XI selection, pitch analysis, opponent assessment, player and tournament development, reporting, and a conversational assistant. Everything is summarized in one web application built using React, TypeScript, Supabase, and powered by local data and fallback features. Coaches can save lots of time as they do not need to switch between separate instruments.

This analytical engine operates in a deterministic way. Players' contributions are expressed in the form of a score with respect to pitch, competition analysis, and contributions of players with the help of direct rules. This is further simplified with match analysis. The XI selection process is conducted according to a greedy approach. The essence of this final description highlights the main asset of the platform, as it shows that every output or recommendation could be traced due to its backstory. The system includes the management aspect of cricket. Refereeing matches, managing tournaments, analysing player strategy, and producing reports are all included in the system, together with providing analysis and granting access to data to specified users. Users can access any information they require through an easy-to-use text assistant.

CrickIQ does not claim accuracy of its predictions. The volume of the data used is relatively low, and the weights are chosen manually before making tests in games that were not used in the dataset. What CrickIQ does is the provision of an open system that can be further improved and supplemented with real-time data and models trained on the dataset that include games not selected for testing.

VIII. FUTURE SCOPE

Presently, CrickIQ has been designed in accordance with T20 format where the matches are shorter and the phases are well defined. Most of the concepts can still be applied to One Day Internationals and Test matches, but the governing principles have to change. For example, ODIs include a longer middle phase where different fielding restrictions are applied, which means that the definitions of phase statistics will have to be changed as well as the role of finisher at the death.

Test match cricket will bring even more changes. The impact score will be required to reward batters who provide durability through the sessions and bowlers who can keep the pressure for a long time, as well as take into account the effect of pitch wear over five days - this cannot be done now, because the static pitch analysis does not consider this factor. The field position module may inform the players how to position their players against batsmen. It operates on your scoring zones, the way you play against types of deliveries involved, and the match situation. CrickIQ contains the statistics based on phases and back matches, which may be used as a starting point. One can present the recommendations on the field map to explain to the captain the reasons for this placement and fulfill the aim of this platform to ensure that all the recommendations have a clear documentary trail.

The player auction obliges teams to assess the value within a given budget, so an auction analysis module may help with this. For example, it may compute the value of the player based on the impact score, position, performance recently, and existing gaps. Also, it may highlight the missing gaps, compare what can be acquired, and track the budget. CrickIQ already points out the problems with team composition, so this new analysis will help with budget issues as well.

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